

sors on rotating components comprise energy harvesting and eliminate battery maintenance.

Electrical actuators in aerospace

EADS will produce 2-seater and 4-seater all-electric aircraft using

rotary electromechanical actuators. Electrical actuators are compact in size, light in weight and function in harsh environments of severe vibration, shock, and high altitudes.

An interesting concept is the use of an additional rear-wheel electric motor drive, which allows E-fan to

taxi on the ground without the main motors and also provides acceleration during take-off.

Flight controls and autopilots

Flight-control systems (FCS) of civil aircraft have undergone huge developments in the last decades. Smart sensors to interface directly to the microprocessors and the fast response actuators to bridge the gap between the logic and the power to be controlled have greatly contributed to the various systems in the aircraft ranging from flight controls to FADEC controls of engine.

Basically, FCS should control the primary and secondary flight control surfaces. The primary control surfaces are pitch, yaw, and roll motions of the aircraft. The secondary flight control surfaces are those which help the aircraft to take off and land like leading-edge slats and spoilers. The horizontal stabiliser at the rear control the pitch motion, the rudder yaw or direction control, and the ailerons for roll motion of the aircraft.

The inertial sensor package sizes are now reduced drastically due to the adoption of micro-electro-mechanical sensor (MEMS) technology. Accelerometers and gyroscopes are known as inertial sensors since they exploit the inertial property of bodies. INS use accelerometers to measure linear acceleration and gyroscopes to sense angular velocity.

The increasing size as well as speed of civil aircraft led to higher control loads. This evolution forced the introduction of hydraulically powered flight-control systems. However, recent advantages in high-performance magnetic materials, miniaturised and highly efficient power electronics, and gear technology make electromechanical actuators (EMAs) increasingly competitive compared to hydraulic actuators regarding power-to-weight ratio, considering the whole flight-control system.

EMAs typically have a permanent magnet electric motor mounted on a reduction gear and/or a linear screw and a power electronics box. Currently, two types of EMAs with linear and rotary output are extensively used in the aircrafts.



Electrical actuators in aerospace (Courtesy: EADS)



Courtesy: Embraer



VH-SJP aircraft (Courtesy: airliners.net)